Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ____________
Date: ____________

Advanced (Habanero)

Q1. Rationalize the denominator: $\frac{1}{\sqrt{2}}$

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{\sqrt{2}}$
(E) $\frac{1}{2}$

Q2. Simplify: $\frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

(A) $\frac{1}{3}$
(B) $\frac{\sqrt{3}}{3}$
(C) $\frac{1}{\sqrt{3}}$
(D) $\frac{3}{\sqrt{3}}$
(E) $\sqrt{3}$

Q3. Rationalize the denominator: $\frac{2}{\sqrt{5}+1}$

(A) $\frac{2(\sqrt{5}-1)}{4}$
(B) $\frac{2(\sqrt{5}+1)}{4}$
(C) $\frac{2(\sqrt{5}-1)}{5-1}$
(D) $\frac{2(\sqrt{5}+1)}{5-1}$
(E) $\frac{2(\sqrt{5}-1)}{5+1}$

Q4. Rationalize the denominator: $\frac{3}{2\sqrt{2}-1}$

(A) $\frac{3(2\sqrt{2}+1)}{7}$
(B) $\frac{3(2\sqrt{2}-1)}{7}$
(C) $\frac{3(2\sqrt{2}+1)}{6}$
(D) $\frac{3(2\sqrt{2}-1)}{6}$
(E) $\frac{3(2\sqrt{2}+1)}{8}$

Q5. Simplify: $(\sqrt{2}+\sqrt{3})^2$

(A) $2+3+2\sqrt{6}$
(B) $2+3-2\sqrt{6}$
(C) $5+2\sqrt{6}$
(D) $5-2\sqrt{6}$
(E) $2+3+\sqrt{6}$

Q6. Rationalize the denominator: $\frac{\sqrt{6}}{\sqrt{2}-\sqrt{3}}$

(A) $\frac{\sqrt{6}(\sqrt{2}+\sqrt{3})}{2-3}$
(B) $\frac{\sqrt{6}(\sqrt{2}-\sqrt{3})}{2-3}$
(C) $\frac{\sqrt{6}(\sqrt{2}+\sqrt{3})}{-1}$
(D) $\frac{\sqrt{6}(\sqrt{2}-\sqrt{3})}{-1}$
(E) $\frac{\sqrt{6}(\sqrt{2}+\sqrt{3})}{2+3}$

Q7. Rationalize the denominator: $\frac{2\sqrt{5}}{\sqrt{5}-2}$

(A) $\frac{2\sqrt{5}(\sqrt{5}+2)}{5-4}$
(B) $\frac{2\sqrt{5}(\sqrt{5}-2)}{5-4}$
(C) $\frac{2\sqrt{5}(\sqrt{5}+2)}{5+4}$
(D) $\frac{2\sqrt{5}(\sqrt{5}-2)}{5+4}$
(E) $\frac{2\sqrt{5}(\sqrt{5}+2)}{5}$

Q8. Simplify: $\frac{1}{\sqrt{7}+1} \cdot \frac{\sqrt{7}-1}{\sqrt{7}-1}$

(A) $\frac{\sqrt{7}-1}{7-1}$
(B) $\frac{\sqrt{7}+1}{7-1}$
(C) $\frac{\sqrt{7}-1}{7+1}$
(D) $\frac{\sqrt{7}+1}{7+1}$
(E) $\frac{\sqrt{7}-1}{6}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify: $\frac{2\sqrt{3}+1}{\sqrt{2}-\sqrt{3}}$

Q10. Simplify the expression: $(\sqrt{2}+\sqrt{5})^2-(\sqrt{2}-\sqrt{5})^2$

Chef's Tips

- Make sure to check the rationalization of denominators carefully.
- Watch for common mistakes such as not simplifying after rationalizing.
- Remind students that rationalizing the denominator does not always simplify the expression.